

Name (print): _____

MYUH# _____

In-class Quiz
#02: Registers and Directives

Directives

1. Write the directive line to reserve an area in memory named `myVar` to save the variables in your code. The area should start at an address that is a multiple of 4 and will store your code.
2. Create a symbol named `myCode` to label the address of the entry point of your code, and let the assembler know the address (be careful about the indents).

Loading registers (move immediate numbers into registers)

3. Write a single assembly instruction to load register **R2** with the immediate hex value 0x93FA.
4. Write a single assembly instruction to load register **R1** with the value currently stored in register **R0**.

ANS: _____

ANS: _____

Arithmetic operations

5. Write the single instruction to subtract the immediate decimal value 2_11010101 from the values currently in **R2** and store the result in register **R1**.
6. Write the single instruction to subtract the values currently in register **R1** from the values currently in **R3** and store the result in register **R3**.

ANS: _____

ANS: _____

7. (5 pts) Write the single instruction to divide the value stored in register **R1** by the value currently in register **R3** and store the result in to register **R0**. (unsigned division)

ANS: _____

Programming (Can use multiple instructions)

8. Use two register **R0** and **R1** to calculate the expression $(9 - 7 \times 12)$, and store the result in register **R0**.
9. (10 pts) Use three register **R0**, **R1** and **R2** to calculate the expression $\frac{2 \times 7 - 12 \times 3 - 10}{9}$, and store the result in register **R0**. (signed division).
10. Use only one register **R0** to calculate the expression $(y - x^2)$, and store the result in register **R0**. Here, y and x can be any small values defined using EQU. (Hint: you can use MVN instruction.) Complete the following code:

```
x    EQU    10
y    EQU    3

    MOV    R0, #x
```